

Multi-Label Retinal Disease Classification

A CNN-Based Medical Image Analysis Project Using the MuReD Dataset

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Dataset: Multi-Label Retinal Diseases (MuReD) Dataset

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GitHub Repository:

<https://github.com/FaisalAhmed21/Multi-Label-Retinal-Disease-Classification>

1. Introduction and Dataset

This project builds a Convolutional Neural Network (CNN)-based medical image analysis model for automatic detection of retinal diseases from fundus photographs. Unlike many classification tasks, a single retinal image can show evidence of more than one condition at the same time (for example, Diabetic Retinopathy alongside Media Haze), so this project is framed as a **multi-label image classification** problem across 20 disease/finding categories, rather than a single-label classification.

The **Multi-Label Retinal Diseases (MuReD)** dataset (Rodriguez Rivera, Al-Marzouqi and Liatsis, 2022) was used. It combines and post-processes images from three established retinal-imaging sources (ARIA, STARE and RFMiD) to provide a higher-quality, better-balanced multi-label benchmark than earlier retinal disease datasets. The dataset contains 2,208 images in total, split into 1,764 training images and 444 validation images, each labelled with 20 binary disease/finding columns (e.g. DR, NORMAL, MH, ODC, ARMD, and others). Images are provided in mixed formats (.png / .tiff) and mixed resolutions (520x520 up to 3400x2800 pixels).

Because MuReD ships with only a training file and a validation file, the validation set was further split 50/50 into a true validation set (222 images, used for model selection and threshold calibration) and a held-out test set (222 images, used only once for final reporting), giving the required train / validation / test structure.

2. Exploratory Data Analysis

All 2,208 image files resolved correctly against their CSV IDs (0 missing files in both train and validation sets). EDA revealed two properties that directly shaped the modelling approach:

- **Severe class imbalance:** DR and NORMAL each have roughly 400 positive examples in training, while several rare classes (CRS, ASR, HTR, CSR) have fewer than 30. This motivated the use of a class-imbalance-aware loss (binary focal cross-entropy) and threshold calibration instead of a plain 0.5 cutoff.
- **Multi-label structure:** many images carry more than one positive label simultaneously, confirming that a sigmoid multi-label output (rather than softmax single-label) is the correct model formulation for this dataset.

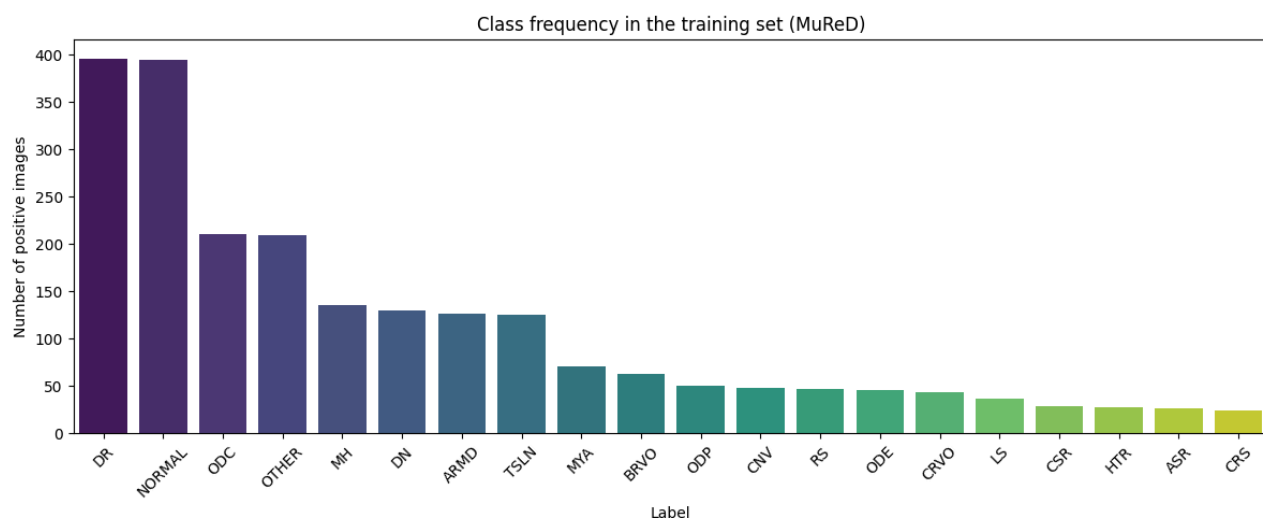


Figure 1. Class frequency in the training set, showing the severe imbalance across the 20 disease labels.

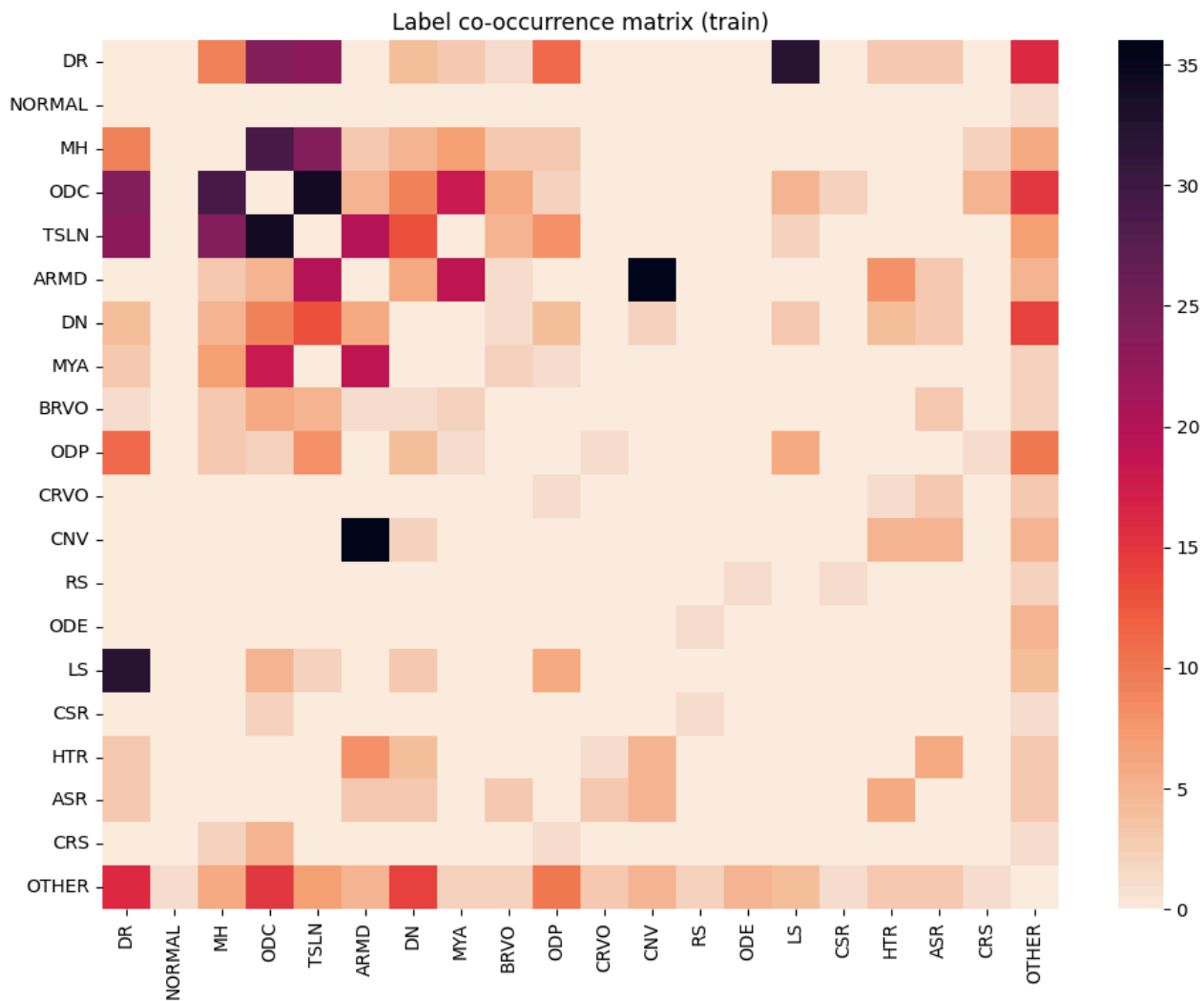


Figure 2: Label co-occurrence matrix, confirming that multiple diseases frequently appear together in the same image.



Figure 3: Sample fundus images from the training set with their ground-truth labels.



Figure 4: Original image resolutions vary substantially (520x520 up to several thousand pixels per side), which is why every image is resized to a common 224x224 input in the preprocessing pipeline.

3. CNN Architecture and Methodology

3.1 Preprocessing

Each image ID was resolved to its actual file (.png or .tiff) on disk, resized to 224x224 (the native input size for the CNN backbones used), and normalized using each backbone's expected preprocessing. Light data augmentation (random horizontal flip, small rotation, zoom and contrast jitter) was applied only to the training set to reduce overfitting, given the relatively small dataset size.

3.2 Model Architecture

Training a CNN from scratch on 1,764 images would overfit badly, so **transfer learning** was used. Three ImageNet-pretrained CNN backbones with different architectural inductive biases were used, each with an identical custom classification head:

- **EfficientNetB0** – compound-scaled efficient convolutions
- **ResNet50** – residual connections
- **DenseNet121** – densely-connected feature reuse

Classification head (identical for all three, trained from scratch): Backbone → GlobalAveragePooling2D → Dropout(0.3) → Dense(256, ReLU) → BatchNormalization → Dropout(0.3) → Dense(20, Sigmoid). A **sigmoid** output (rather than softmax) was used because each of the 20 labels is an independent binary decision — several can be simultaneously true for one image.

The predictions of the three backbones were combined using **soft-voting ensembling** (averaging the predicted probabilities of all three models).

3.3 Training Process

Each backbone was trained in two phases, a standard and effective transfer-learning recipe:

- **Phase 1 (warm-up):** backbone frozen, only the new classification head trained for up to 10 epochs at learning rate 1e-3, so the randomly-initialized head does not destroy the pretrained features.
- **Phase 2 (fine-tuning):** the backbone is unfrozen and trained end-to-end for up to 10 more epochs at a much lower learning rate (1e-5), letting the network adapt its high-level visual features specifically to retinal images.

Loss function: Binary Focal Cross-Entropy (with class balancing, gamma = 2.0) was used instead of plain binary cross-entropy, to down-weight easy/majority-class examples and focus learning on the harder, minority-class examples given the severe imbalance identified in the EDA.

Callbacks: Early stopping (patience 6, monitoring validation AUROC), learning-rate reduction on plateau, and model checkpointing (saved to Google Drive so training progress survives Colab session disconnects) were used for all three backbones.

Decision thresholds: rather than using a fixed 0.5 cutoff for every class, per-class thresholds were selected on the validation set using Youden's J statistic. However, because several classes had fewer than 10 positive/negative validation examples (making this statistic unreliable / noise-fitting for those classes), a minimum-support guard was added: classes with insufficient validation support fall back to the standard 0.5 threshold.

4. Training and Evaluation Results

4.1 Training Curves

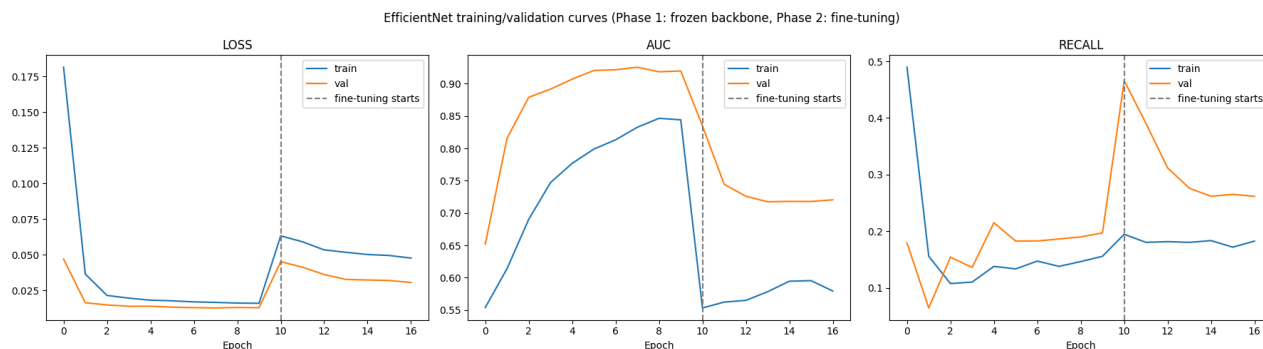


Figure 5. EfficientNetB0 training/validation loss, AUROC and recall over epochs. The dashed vertical line marks the transition from Phase 1 (frozen backbone) to Phase 2 (fine-tuning).

4.2 Single-Model vs. Ensemble — Was the Ensemble Worth It?

To justify the added training cost of using three backbones instead of one, EfficientNetB0 alone was evaluated against the full 3-model ensemble on the held-out test set:

Metric	EfficientNet Only	3-Model Ensemble
Macro-F1	0.2637	0.2542
Micro-F1	0.3343	0.4596
Hamming Loss (lower is better)	0.1579	0.0980
Macro-AUROC	0.8268	0.8944

The ensemble improved overall discrimination (Macro-AUROC +8%) and reduced the fraction of

individually wrong label predictions (Hamming loss, Micro-F1). However, Macro-F1 — the metric that weights all 20 classes equally, including the rarest ones — did not improve. This indicates the ensemble helps on classes that already had reasonable support, but does not solve the underlying data-scarcity problem for the rarest classes (as few as 3–8 test examples), which is a limitation of dataset size rather than model architecture.

4.3 Per-Class Test Set Results (Ensemble)

Class	Precision	Recall	F1	Support
DR	0.64	0.86	0.73	49
NORMAL	0.50	0.96	0.66	49
MH	0.37	0.65	0.47	17
ODC	0.27	0.88	0.42	26
TSLN	0.37	0.94	0.53	16
ARMD	0.18	0.88	0.29	16

DN	0.15	0.44	0.22	16
MYA	1.00	0.44	0.62	9
BRVO	0.00	0.00	0.00	8
ODP	0.00	0.00	0.00	6
CRVO	0.00	0.00	0.00	5
CNV	0.00	0.00	0.00	6
RS	1.00	0.17	0.29	6
ODE	0.67	0.40	0.50	5
LS	0.00	0.00	0.00	5
CSR	0.00	0.00	0.00	3
HTR	0.00	0.00	0.00	4
ASR	0.00	0.00	0.00	3
CRS	0.00	0.00	0.00	3
OTHER	0.25	0.73	0.37	26
Macro avg	0.27	0.37	0.25	278

Micro avg	0.35	0.67	0.46	278
Weighted avg	0.38	0.67	0.44	278

Nine of the twenty classes (BRVO, ODP, CRVO, CNV, LS, CSR, HTR, ASR) score 0.00 F1 on the test set. Each has fewer than 10 test examples; with the minimum-support guard in place these classes fall back to a conservative 0.5 threshold, under which the model does not confidently predict them positive. This is discussed further in Section 6.

5. Sample Predictions and Model Interpretability

To verify that the model bases its predictions on genuine retinal features rather than background artifacts, **Grad-CAM (Gradient-weighted Class Activation Mapping)** was used to visualize which regions of each test image most influenced the model's prediction.

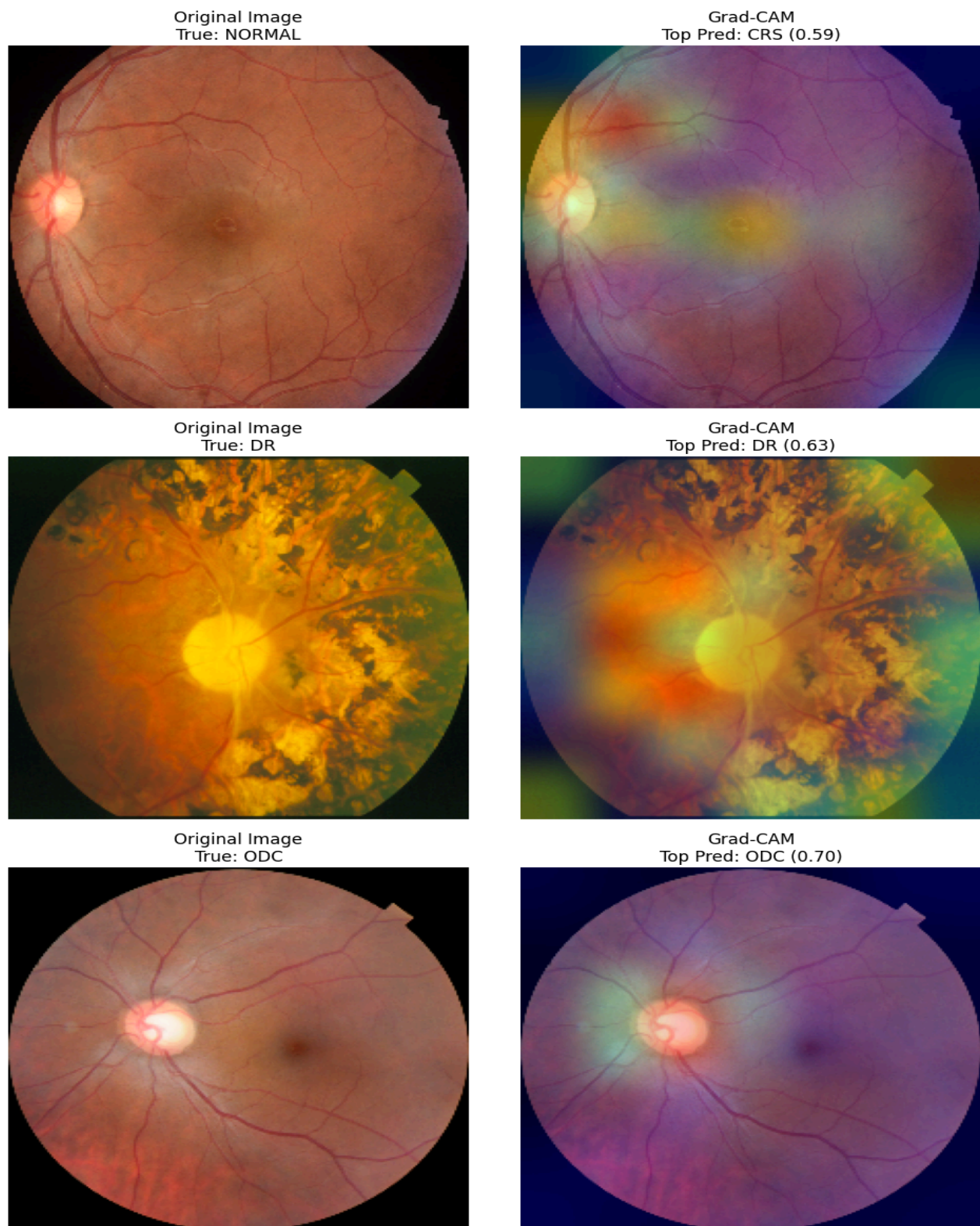


Figure 6: Grad-CAM visualizations on sample test images: original image (left) vs. the model's attention heatmap overlaid with its top predicted label and confidence (right).

6. Discussion, Limitations and Conclusion

This project demonstrates a complete, correctly-formulated multi-label medical image classification pipeline: transfer learning with three CNN backbones, a focal loss suited to severe class imbalance, calibrated decision thresholds, and Grad-CAM interpretability. The ensemble achieved a Macro-AUROC of 0.894, a genuinely strong, threshold-independent result for a 20-class imbalanced multi-label problem.

The key limitation, made explicit by the single-model-vs-ensemble ablation in Section 4.2, is that **the bottleneck is dataset size for rare classes, not model architecture**. Several disease labels have fewer than 10 examples in the entire test set; no amount of additional backbone diversity can manufacture signal that is not present in that few training examples. The 3-model ensemble measurably improved aggregate metrics (AUROC, Micro-F1, Hamming loss) but did not move Macro-F1, confirming that its benefit is concentrated on classes that already had reasonable support.

Future work that would most directly address this limitation:

- Targeted oversampling or synthetic augmentation specifically for the rarest classes (BRVO, ODP, CRVO, CNV, LS, CSR, HTR, ASR, CRS)
- Collecting or sourcing additional labelled examples for those classes specifically, rather than further architectural changes
- Per-class threshold tuning with a more robust small-sample estimator (e.g. cross-validated thresholds) instead of a single validation split

GitHub Repository: <https://github.com/FaisalAhmed21/Multi-Label-Retinal-Disease-Classification> - contains the complete Jupyter Notebook, this report, and the dataset CSVs.